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Abstract

Optimizing atomic layer deposition (ALD) process requires precise control of multiple parameters, whose nonlinear interactions often demand extensive experimental iterations. Here, we present a machine learning (ML) framework for developing ALD model that accelerates process optimization. The framework is applied to predict hafnium oxide (HfO_x) film properties—thickness (T_{ox}), refractive index (RI), and wet etch rate (WER)—from ALD process parameters. Trained on 215 experimental data points using a deep-neural network (DNN), the model achieves prediction accuracy above 90% within ranges of ± 1.0 nm for T_{ox} , ± 0.04 for RI, and ± 0.9 Å/min for WER. Beyond numerical prediction, the framework generates film-property maps that enable process visualization and support autonomous ALD optimization. Comparable performance is retained with 100 training data points, effectively capturing nonlinear trends and extending predictability beyond experimentally measured ranges. This approach on ML-driven ALD model significantly reduces experimental time and resource consumption while effectively addressing complex multi-parameter challenges in ALD. Importantly, these results lay the groundwork for integrating ML into ALD digital twin platforms, establishing a foundation for next-generation autonomous semiconductor process development.

Keywords

: Machine learning (ML); deep neural network (DNN); atomic layer deposition (ALD); film quality prediction; digital twin

In the digital era, artificial intelligence (AI) is driving the development of efficient and innovative technological systems. Industry 4.0 integrates ‘intelligent’ information technology into manufacturing and industrial processes, with machine learning (ML) offering the potential to enhance flexibility, improve efficiency, and minimize process errors. This transformative technology is expected to influence virtually every sector, reshaping systems and protocols across a wide range of fields.^{[1]–[3]}

In semiconductor fabrication, the integration of ML is particularly valuable as processes grow increasingly complex, requiring tight design footprints and intricate three-dimensional (3D) structures.^{[5],[6]} Process optimization is also critical for achieving high-quality films and simplifying process steps. Establishing proof-of-concept for ML-driven thin film fabrication in semiconductor devices is thus essential to demonstrate its potential for advancing innovative technical approaches. Previous studies have explored ML-driven techniques in various thin film fabrication processes, including atomic layer deposition (ALD),^{[7]–[9]} chemical vapor deposition (CVD),^{[10]–[13]} sputtering,^{[14]–[16]} pulsed laser deposition (PLD),^[17] chemical-mechanical planarization (CMP),^[18] and etching.^{[19],[20]}

Among these techniques, ALD stands out as a promising technique for emerging semiconductor devices due to its ability to deposit highly conformal films. Its primary advantages are self-limiting growth mechanism and precise control of angstrom-level film thickness.^[21] These features enable the fabrication of complex 3D device architectures. Continued process optimization and the discovery of new materials are essential to improving film quality and broadening ALD’s applicability in cutting-edge semiconductor manufacturing. Optimizing ALD processes involves careful control of multiple parameters, including precursor pulse and purge times, chamber and precursor temperatures, as well as additional factors such as working pressure and pumping capacity.^{[22]–[24]}

The comprehensive exploration of all possible ALD process conditions, however, is economically unfeasible, as each experimental run costs over a thousand dollars per wafer, plus additional labor and analysis costs are also a big burden.^{[2],[5]} This makes the incorporation of AI technologies particularly promising in improving cost-efficiency of process development by minimizing the volume of experimental data required.

In this study, we present an ML-driven ALD framework utilizing deep neural networks (DNNs),

with a focus on deposition for hafnium oxide (HfO_x) films, toward ALD digital twin. While conventional ML applications in ALD have shown promise across various algorithms in predictive tasks, they have primarily been limited to monitoring film thickness. Although film thickness is often used to verify saturation behavior within the ALD window, it provides only a partial understanding of film quality. To address this limitation and extend the scope, we incorporate film density evaluation, an essential factor in assessing durability and etch resistance for qualitative ranking of film properties. Specifically, we investigate film density using wet etch rate (WER) testing, which serves as an indicator of the atomic packing density in HfO_x films. Higher-density films can resist etchant penetration, resulting in slower etching rates, whereas lower density films exhibit the opposite behavior.^{[25],[26]}

Within this scope, this work focuses on an ALD development framework that uses existing tool signals with selected characterization data to learn film-property correlations. The work is therefore intended as an initial proof-of-concept for data-efficient ALD process development in the current reactor environment. Additionally, we highlight the unique strengths of NNs in regression tasks, particularly their ability to capture intricate, non-linear dependencies in ALD processes. Such models not only deliver precise predictions of film properties but also generates prediction maps that guide process optimization and reveal parameter interdependencies, beyond the conventional approaches limited to prediction-percentage analysis. By integrating ML-based prediction into the ALD workflow, this study marks experimental design and illustrates how these innovations are reshaping methodologies for process optimization.

Results

Framework for ML-driven ALD modeling

The HfO_x ALD processes were investigated by using tetrakis(dimethylamido)hafnium (TDMA-Hf) as Hf-precursor and water (H_2O) as oxidant. Data mining for DNN training was conducted by varying three ALD conditions: deposition temperature, precursor temperature (TDMA-Hf), and sample location among various ALD parameters (**Fig. 1a**). Other ALD parameters, such as precursor pulse/purge time and number of cycles etc., were kept consistent across all data points in this study. The deposition temperature significantly influences reaction thermodynamics and kinetics for the overall deposition process, as defined by the ALD windows.^[27] At lower

temperatures, incomplete reactions and/or precursor condensation occur, while higher temperatures lead to precursor decomposition and/or desorption. The optimal deposition temperature, therefore, ensures self-limiting surface reactions, a key factor for ALD. Meanwhile, the precursor temperature affects the vapor pressure of TDMA-Hf, determining proper vaporization with sufficient supply to the substrate surface. This impacts the film uniformity during the ALD process. The uniformity was assessed at nine points in a 4-inch wafer, with coordinates measured from the center (0, 0) at the inlet side. This location parameter helps evaluate how growth reactions vary across the wafer surface, reflecting precursor flow/reaction dynamics and overall process uniformity.

The properties of HfO_x film, such as film thickness (T_{ox}), refractive index (RI), and WER, were then evaluated. All these listed film properties were used for labels in training and testing a DNN model, as presented in **Fig. 1b**. A DNN model was employed because it provides a flexible framework for learning nonlinear relationships between ALD conditions and multiple correlated film properties within a unified multi-output model. The developed DNN architecture consists of four input neurons (representing ALD conditions) and three output neurons (representing film properties) with multiple hidden layers. The multiple hidden layers help to capture complex and non-linear relationships in ALD processes, which generates more precise feature extraction compared to single layer perceptron.^{[28],[29]}

The training and testing algorithm of the DNN system is illustrated in **Fig. 1c**. Data collection yielded 315 data points: 215 for training, and 100 for testing (see **Supplementary Fig. 1** for all 315 data points obtained). For training and testing, all values of process parameters and film properties were normalized to be on a comparable scale. The NN output was calculated using the following vector-matrix multiplication equation:

$$y_j = \sum_{i=1}^4 w_{i-j} \cdot x_i + b_{m+1} \quad (i = 1,2,3,4 \text{ \& } j = 1,2,3 \text{ \& } m \text{ is \# of last hidden layers})$$

where y is output, x is input, b is bias, and w represents synaptic weights indicating coefficients and independent terms of the linear transformation. These y values are then activated by an exponential linear unit (ELU) function to generate the NN output. The ELU activation function was adopted in the present model because its smooth nonlinear response and stable gradient behavior are advantageous for learning nonlinear ALD process-property relationships. Finally, the calculated NN output is compared with the desired output, predictions are made, and error

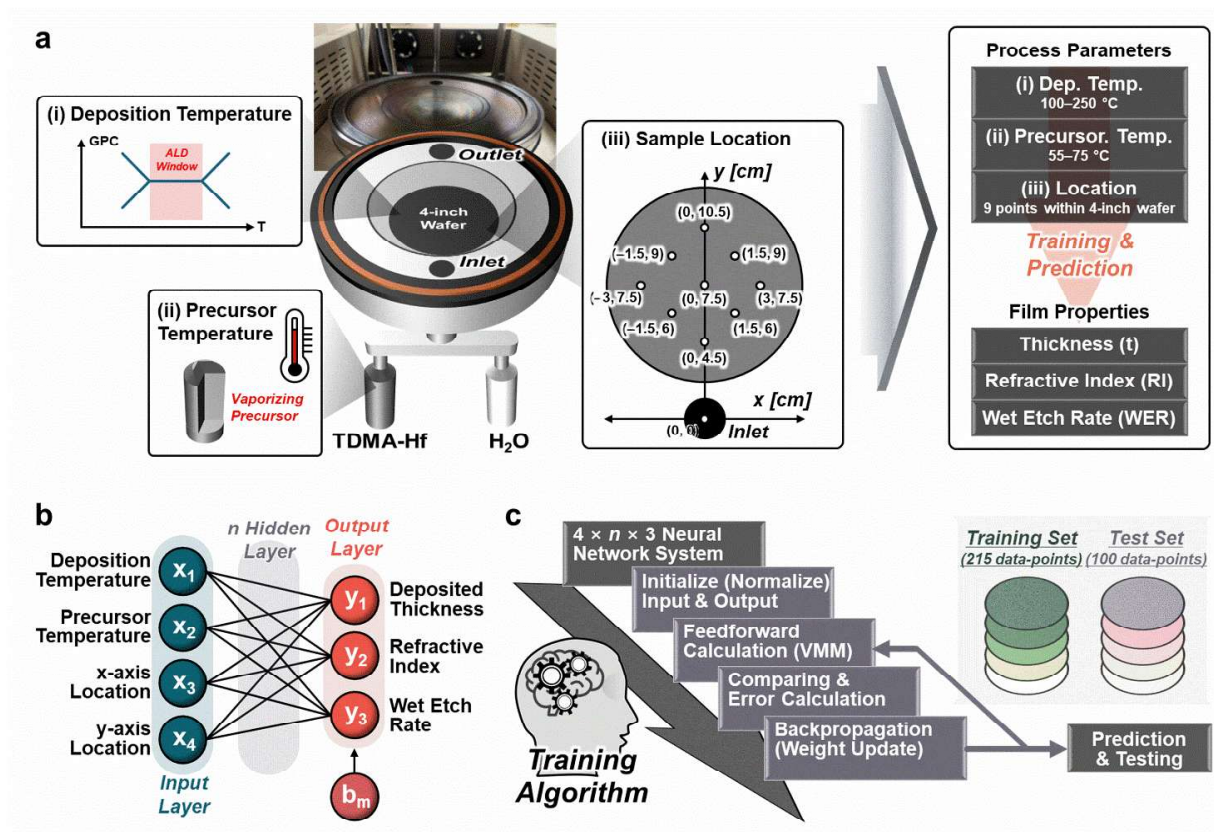


Fig. 1 | Schematics of the ALD process and deep neural network (DNN) framework for HfO_x film deposition. **a** Controlled ALD process parameters for DNN-driven optimization: (i) deposition temperature, (ii) precursor temperature for TDMA-Hf, and (iii) sample location coordinates on a 4-inch wafer. The process parameters and film properties of interest are displayed on the right. **b** DNN architecture with 4 input neurons representing ALD parameters and 3 output neurons for film properties. **c** Training and testing algorithm for the multi-layer NN system, utilizing 215 data points for training and 100 data points for testing.

calculation is performed through backpropagation with weight updates.

Experimental setup for ALD modeling

The detailed ALD process conditions and DNN systems are illustrated in **Fig. 2**. The ALD process operated at a constant working pressure of ~ 310 mTorr, with argon as the carrier gas flowing at 40 sccm (**Fig. 2a**). Each deposition cycle followed a fixed pulse sequence: 0.2/20-sec pulse/purge time for both TDMA-Hf and H_2O precursors. This cycle was repeated 100 times for each experimental run. While these parameters remained constant, the deposition temperature was investigated ranging from 100°C to 250°C in steps of 25°C . At the same time, the TDMA-Hf precursor temperature was varied between 55 – 75°C with 5°C steps. This systematic approach to parameter variation facilitated a thorough exploration of the ALD HfO_x process, generating comprehensive data points for DNN training and operation.

Figure 2b illustrates the detailed DNN architecture employed for predicting HfO_x film properties. The input layer comprises four neurons corresponding to ALD process parameters: deposition temperature, precursor temperature, and x- and y-axis locations on the wafer. These inputs are fed through five hidden layers with decreasing neuron counts ($128 \times 64 \times 32 \times 16 \times 8$). The decreasing neuron counts are selected to prevent the distortion of net output values in output layer. The output layer consists of three neurons, representing the film properties of interest: T_{ox} , RI, and WER. Bias terms (b_1 to b_6) are incorporated at each layer to enhance the flexibility and predictive accuracy of DNN model.

Figure 2c describes experimental methods to measure three film properties. Spectroscopic ellipsometry tool was utilized to determine T_{ox} and RI of the HfO_x films. The analysis was performed with three spectra at different incident angles (55° , 65° , 75°), using the Cauchy model to calculate the HfO_x film thickness. For the WER test, initial film thickness was measured, followed by etching in a 500:1 hydrofluoric acid (HF) solution for either 1 or 5 minutes, depending on the initial thickness (1-min etching for films below 8 nm, 5-min etching for thicker films). The remaining thickness was measured and compared with the initial thickness to estimate the WER. These measurements of T_{ox} , RI, and WER were performed across all temperature and location variations for ALD, generating extensive data points for DNN training and testing.

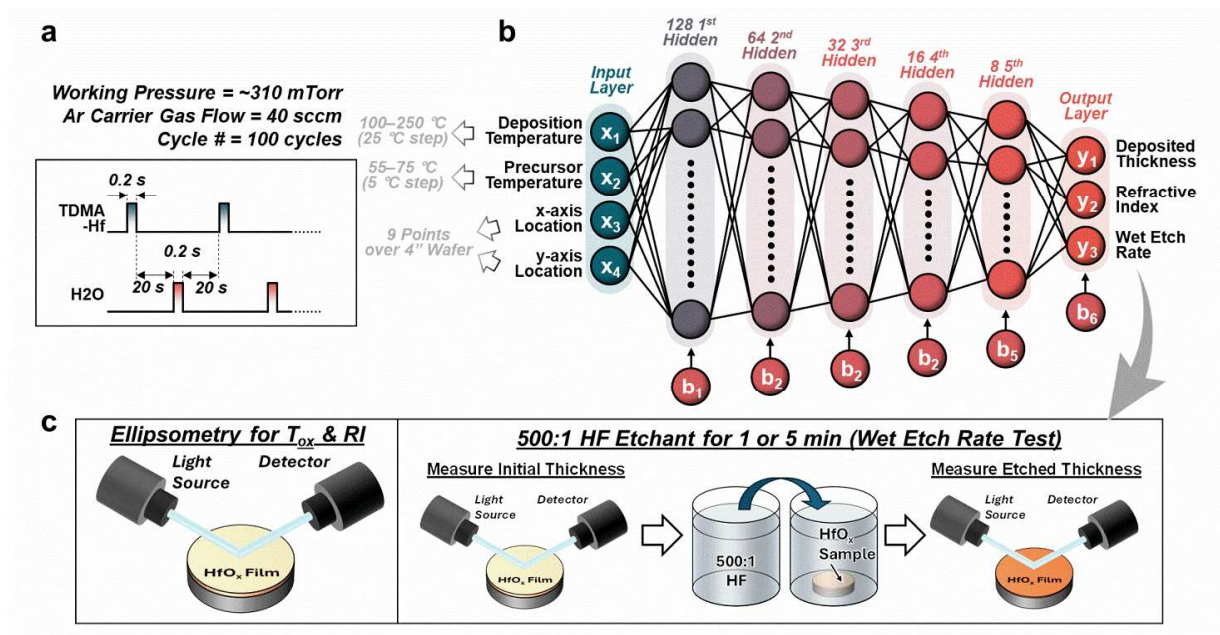


Fig. 2 | Detailed ALD process conditions and experimental methods for data collection. **a** Various ALD process parameters including working pressure, carrier gas flow, cycle number, and pulse/purge sequence for TDMA-Hf and H₂O precursors. Variable parameters include deposition temperature (100–250 °C), precursor temperature (55–75 °C), and sample location (nine points over 4" wafer). **b** Optimized DNN architecture with 4 input neurons, 5 hidden layers (128, 64, 32, 16, and 8 neurons), and 3 output neurons. Bias terms (b_1 to b_6) are included at each layer. **c** Experimental methods for measuring film properties: ellipsometry for thickness (T_{ox}) and refractive index (RI), and wet etch rate (WER) test using 500:1 HF etchant for 1 or 5 minutes to determine film density.

Impact of multi-dimensional ALD conditions

Figure 3 illustrates why AI/ML is valuable for ALD process optimization. When a single ALD parameter is varied, clear and interpretable trends in film properties often emerge (**Fig. 3a**). For example, increasing the deposition temperature, while holding the precursor temperature at 75 °C and the sample location at (0 cm, 7.5 cm), reduces T_{ox} and WER and increases RI. This behavior is consistent with more complete surface reactions at higher temperatures; at lower temperatures, unreacted sites persist and the film becomes less dense.^{[30],[31]} Likewise, when the TDMA-Hf precursor temperature is raised at a fixed deposition temperature of 100 °C and the same sample location, film properties stabilize once the precursor temperature exceeds ~60 °C, reflecting the precursor's vaporization threshold; below this point, insufficient vapor supply limits effective ALD cycling. In such single parameter sweeps, optimization is relatively straightforward because the trends are monotonic and mechanistically traceable.

By contrast, optimization becomes more challenging when considering two variables simultaneously. **Figure 3b** varies deposition temperature and sample position concurrently (with the precursor temperature fixed at 55 °C), revealing strong spatial non-uniformity at lower temperatures (100–175 °C): T_{ox} is higher and RI/WER are lower near the inlet across the 4-inch wafer. As the deposition temperature increases, the film becomes more uniform, with optimal uniformity achieved above ~225 °C. These experimental results reflect coupled transport-reaction effects: at low deposition temperature, the process is reaction-limited and therefore highly sensitive to local precursor delivery and/or residence time, resulting in non-linear responses across position and temperature.

To properly investigate these trade-offs, multidimensional parameter control is essential. One-factor-at-a-time sweeps are insufficient, because varying a single parameter cannot capture the interdependencies and trade-offs across film-quality metrics. This complexity motivates ML integration: data-driven regression can capture the coupled response surface and generate continuous prediction maps, enabling multi-objective optimization and systematic exploration of parameter interdependencies, rather than relying on isolated one-dimensional sweeps.

ALD model prediction accuracy

Figure 4 illustrates the DNN prediction accuracy for ALD film-quality under different

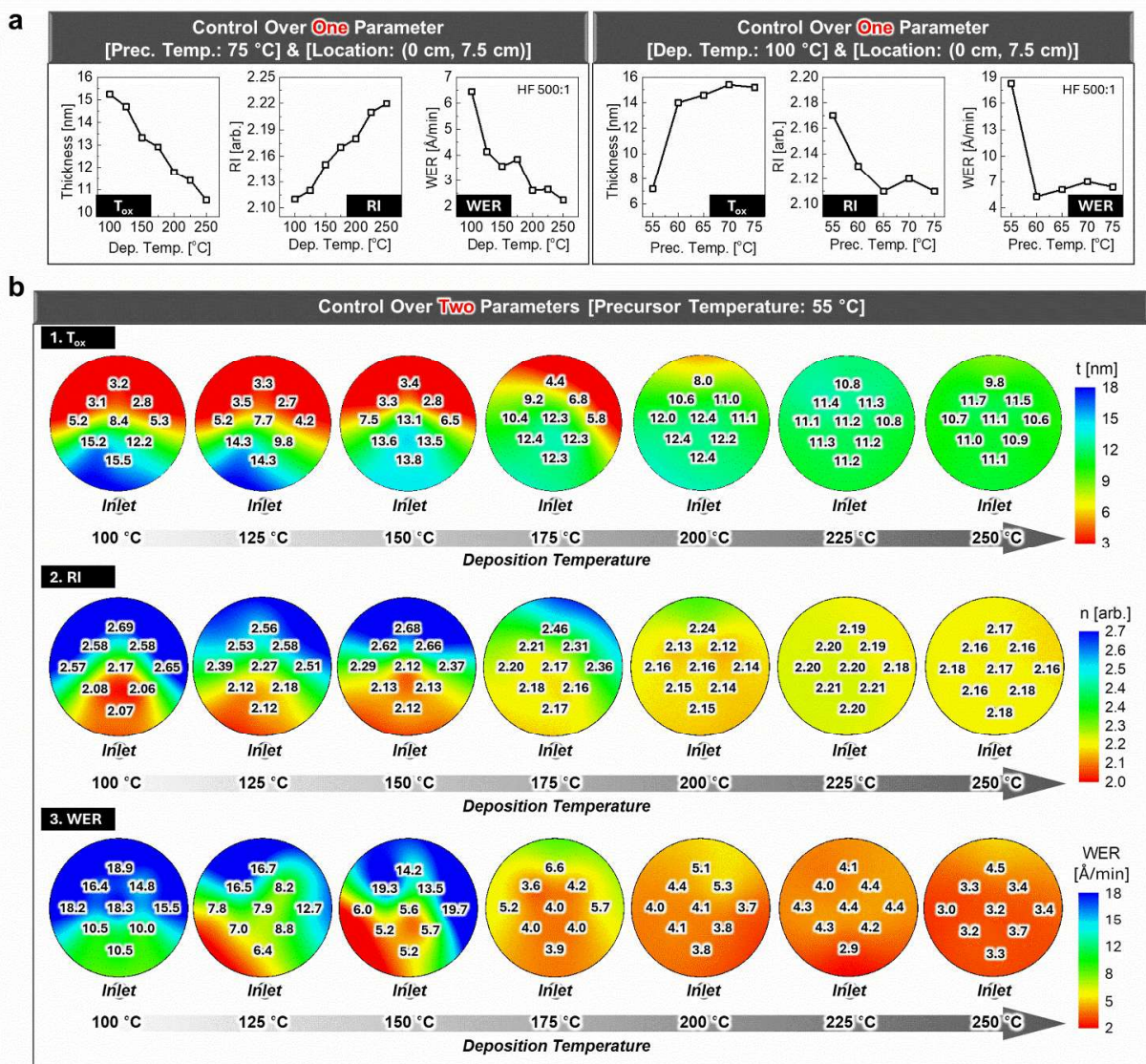


Fig. 3 | Effects of ALD parameters on HfO_x film properties. **a** Single parameter control: left panels show the impact of deposition temperature on T_{ox} , RI, and WER at fixed precursor temperature (75 °C) and location (0 cm, 7.5 cm). Right panels demonstrate the effect of precursor temperature at fixed deposition temperature (100 °C) and location. **b** Two-parameters control: Wafer maps showing the distribution of T_{ox} , RI, and WER across various deposition temperatures (100–250 °C) at a fixed precursor temperature (55 °C). These maps illustrate complex interactions between multi-dimensional ALD parameters, reflecting the non-linear trends of ALD processes.

validation scenarios. To assess the model's capability, the dataset was randomly partitioned into two subsets: set X -1 (light blue, 215 data points) and set X -2 (dark blue, 100 data points), allowing for cross-validation of the DNN performance (where X denotes the distribution set number) (**Fig. 4a**).

When the model was both trained and tested on the same subset (training accuracy, using X -1 only), the accuracy approached 100%, reflecting that the DNN successfully captured the data trends. More importantly, cross-validation using X -1 for training and X -2 for testing (test accuracy) yielded strong predictive performance across all film metrics: 92–95% for T_{ox} , 95–97% for RI, and 90–95% for WER (see **Supplementary Fig. 2** for hidden-layer optimization). Model reliability was further evaluated by three-fold cross-validation, as shown in **Supplementary Fig. 3**. The consistently high test accuracy across three independent dataset partitions suggests that no severe overfitting was observed in the present model. These results demonstrate that the model not only fits the training data but also generalizes reliably to unseen process conditions, highlighting its suitability for ALD process optimization.

Figures 4b–d further evaluate model performance by varying the prediction tolerance for each property. Here, the prediction range denotes the acceptable error window within which the predicted value is considered correct, *i.e.*, $|y_{\text{prediction}} - y_{\text{label}}| \leq \delta$ (*prediction range*). Training accuracy remained nearly perfect (~100%) across strict tolerances, while test accuracy degraded gradually as the ranges narrowed. Maintaining accuracy above 90% requires prediction windows of approximately ± 1.0 nm for T_{ox} , ± 0.04 for RI, and ± 0.9 Å/min for WER. As expected for supervised learning, the availability of more training data would likely improve these limits. Overall, these results demonstrate that the DNN framework can accurately predict HfO_x film properties across a wide range of deposition conditions and narrow prediction windows, confirming its promise for ALD process modeling and optimization.

ALD prediction maps

To assess the ALD model's predictive scope, we compared DNN-generated film-property maps (T_{ox} , RI, and WER) with experimental data. To assess the ALD model's predictive scope, we compared DNN-generated film-property maps (T_{ox} , RI, and WER) with experimental data. The analysis focuses on a representative wafer location at (0 cm, 10.5 cm), where the output variation

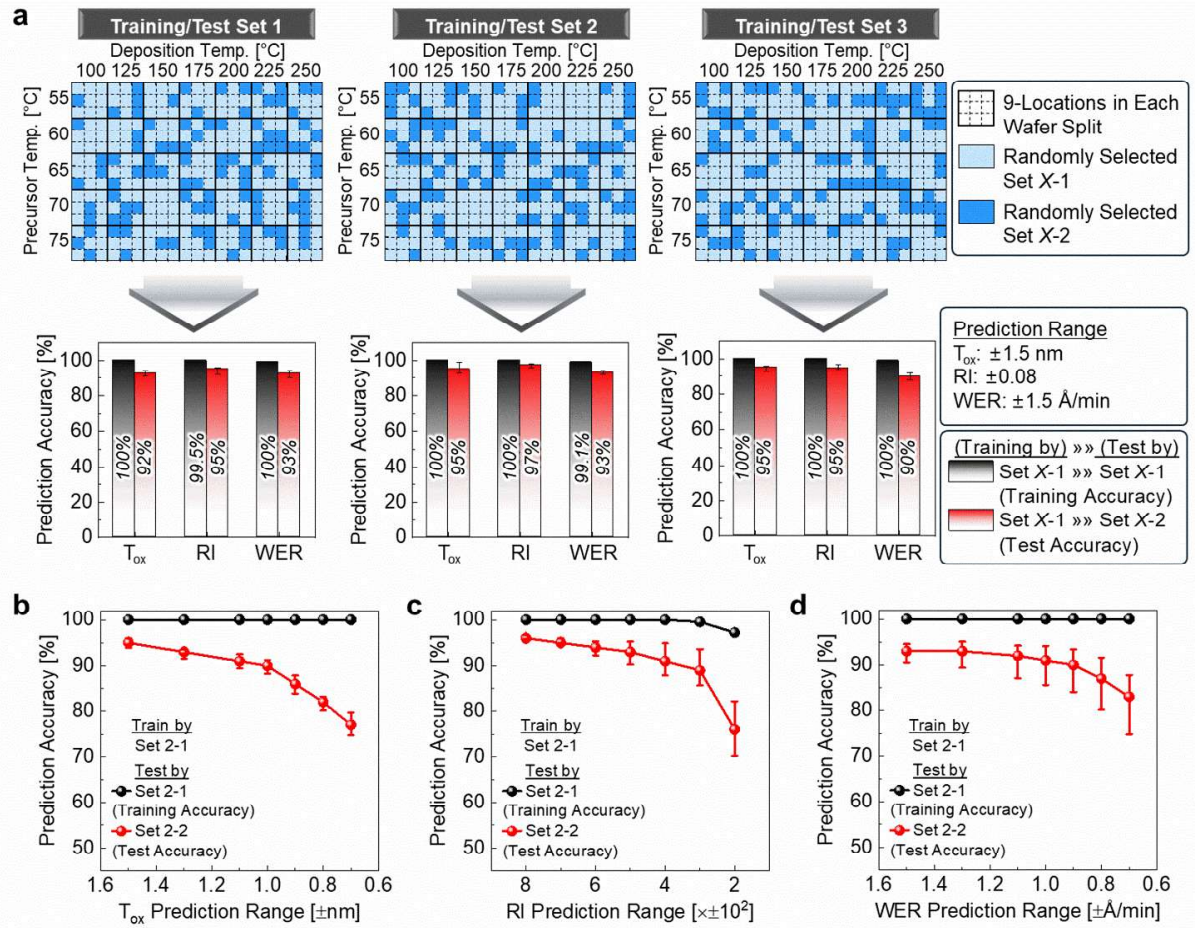


Fig. 4 | Prediction accuracy for ALD process from the training/testing variations on DNN system. **a** 2D maps of three training/testing sets, with corresponding prediction accuracies for film properties. For the test, the prediction ranges of ± 1.5 nm, ± 0.08 , and ± 1.5 Å/min were used for T_{ox} , RI, and WER, respectively. **b–d** Prediction accuracy versus prediction range for **b** T_{ox} , **c** RI, and **d** WER, comparing training (tested with set 1-1) and prediction (tested with set 1-2) accuracy. These accuracies were obtained after 1000 training epochs and are reported as mean values with error ranges derived from 10 independent training/testing runs.

across precursor and deposition temperature space was pronounced. **Figure 5a** presents the experimental property maps as functions of precursor temperature and deposition temperature, serving as the ground truth. Using training sets of varying size (100–215 data points), prediction maps were generated (**Fig. 5b**).

As shown in **Figs. 5c–e**, the predicted surfaces reproduce the overall shapes and gradients of the experimental maps across training set sizes. These trends are consistent with known ALD behavior: at lower precursor temperatures, reduced precursor vapor pressure can limit precursor supply, leading to lower T_{ox} and higher WER owing to reduced film density. In addition, at lower deposition temperatures, increased physisorption of precursor ligands can enhance apparent film growth while yielding less dense films, again resulting in elevated WER. Quantitative accuracy, measured within prediction windows of ± 1.5 nm (T_{ox}), ± 0.08 (RI), and ± 1.5 Å/min (WER), declines somewhat when only 100 points are used (see **Supplementary Fig. 4** for robustness analysis under assumed process variation). Nevertheless, DNN preserves the key qualitative dependencies of ALD conditions on film properties in prediction maps. This robustness demonstrates that regression models can capture essential process trends even under limited data availability, highlighting strong qualitative agreement with experiments even under reduced training data. Therefore, the ML integration can guide experimental design, prioritize critical process windows, and reduce the number of costly and time-consuming trials required for ALD process optimization.

Extrapolation beyond experimental ranges

Building on the validation in Fig. 5, we next examine the model's predictive scope beyond the measured parameter space. **Figure 6** compares experimental and predicted 3D maps of T_{ox} , RI, and WER across multiple sample locations. The experimental maps (dark surfaces) span precursor temperatures of 55–75 °C and deposition temperatures of 100–250 °C. In contrast, the predicted maps (colored surfaces) extend these ranges to 50–80 °C for precursor temperature and 75–275 °C for deposition temperature, based on training with 100 data points (**Supplementary Fig. 5**). This extrapolation demonstrates a key advantage of ML-based ALD predictions: the ability to forecast film properties beyond the experimental dataset (see **Supplementary Fig. 6** for maps across all sample locations). A representative case is the T_{ox} map at $(x, y) = (0, 4.5)$. While experiments show

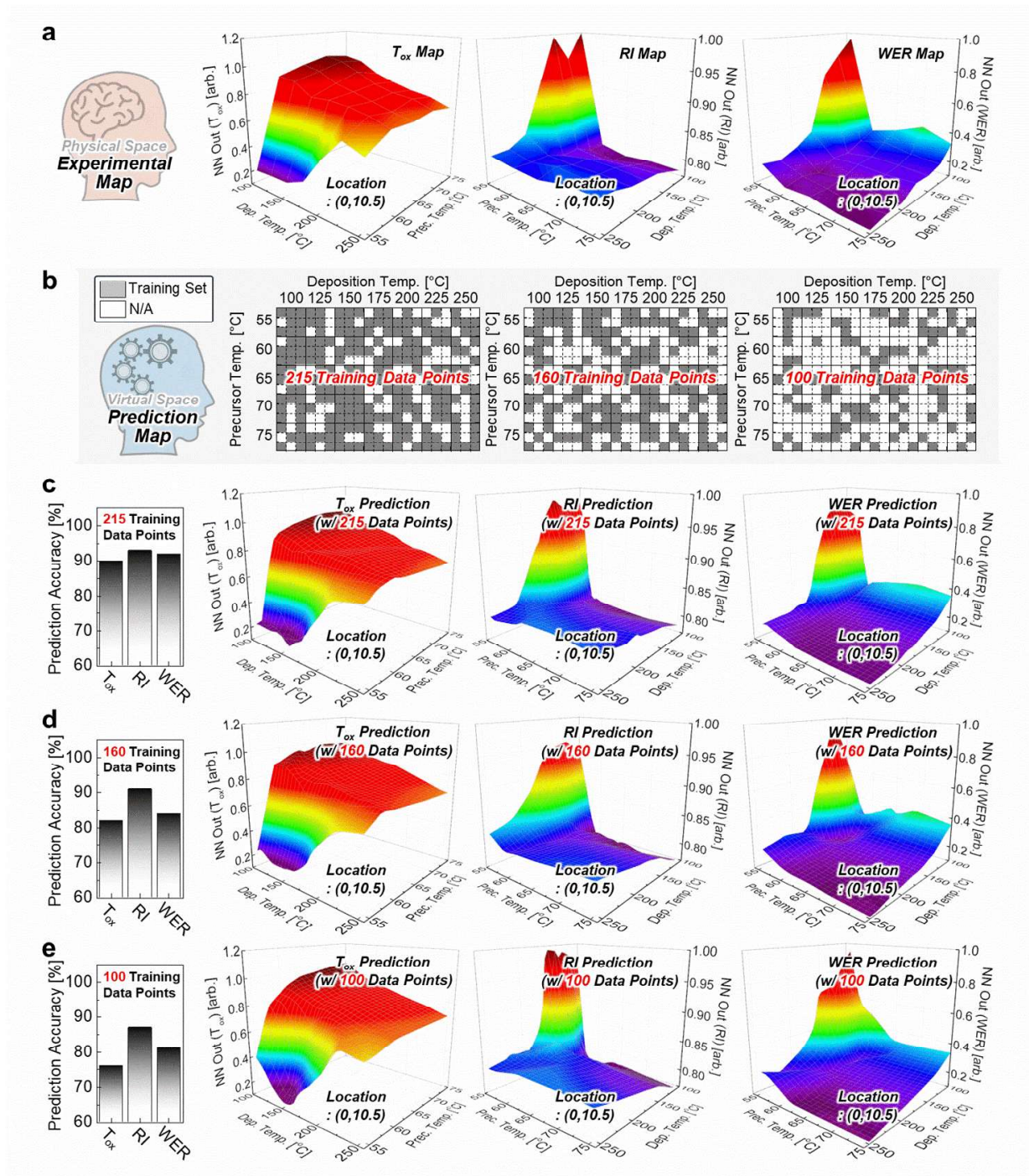


Fig. 5 | DNN-based prediction maps for ALD HfO_x film properties. **a** Experimental maps of T_{ox}, RI, and WER at location (0 cm, 10.5 cm). **b** Training set configurations with different dataset sizes. **c–e** DNN prediction maps using **c** 215, **d** 160, and **e** 100 training data points. Each row

shows prediction accuracy bar graphs and 3D prediction maps for T_{ox} , RI, and WER at location (0 cm, 10.5 cm).

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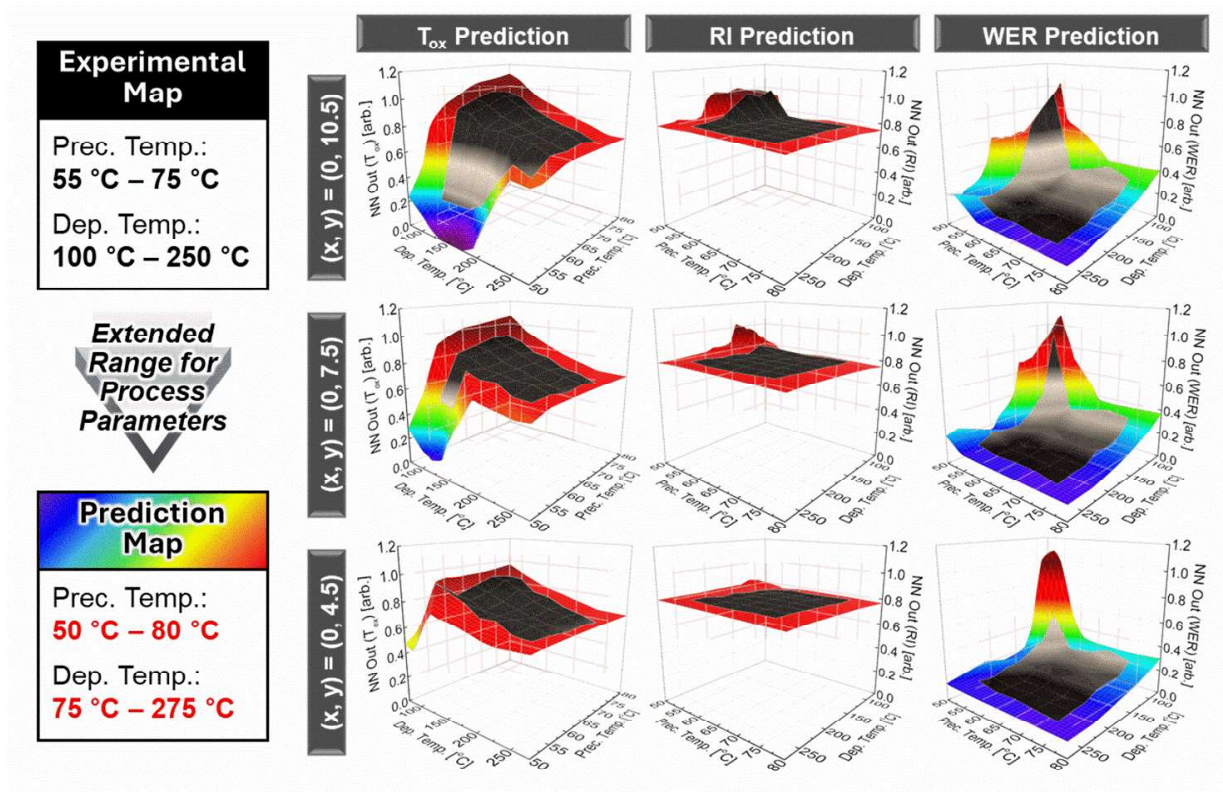


Fig. 6 | 3D surface plots showing experimental and prediction maps for three film properties: T_{ox}, RI, and WER. The plots are arranged in three rows representing different sample locations $(x, y) = (0, 10.5)$, $(0, 7.5)$, and $(0, 4.5)$. Each column corresponds to a specific property prediction (T_{ox}, RI, WER). The experimental map (black-colored plots) covers precursor temperatures of 55 °C–75 °C and deposition temperatures of 100 °C–250 °C. The prediction maps (colored plots) extend these ranges to 50 °C–80 °C for precursor temperature and 75 °C–275 °C for deposition temperature, demonstrating the capability of the ML-based model to predict film properties over an extended range of process parameters.

a monotonic increase as deposition and precursor temperatures decrease, the DNN predicts that T_{ox} would ultimately decline further if the parameters extended beyond the measured window (**Supplementary Fig. 7**). This discrepancy may reflect the fact that the DNN learns global correlations across the full dataset, including other sample locations and parameter combinations, rather than merely extending the local trend observed at a single condition.

However, because no explicit physics-based constraint was imposed, predictions outside the experimental domain should be interpreted cautiously and used only as exploratory guidance. For example, precursor temperatures as low as 50 °C are physically unrealistic for HfO_x deposition because the vapor pressure of TDMA-Hf is insufficient to sustain film growth, as supported by the no-growth behavior shown in **Supplementary Fig. 8**. These results indicate that ML-guided process optimization is most reliable within expert-defined process windows, and that inclusion of failure-region data can further improve model robustness by enabling the model to learn both valid and invalid process regimes. Overall, while additional experiments would refine quantitative accuracy in extended ranges, these findings highlight that ML-driven ALD modeling can reproduce physically meaningful process-dependent trends, including ALD window behavior and failure boundaries. Such advanced predictive capability offers substantial reductions in time and resources for digital-twin-based process development in semiconductor manufacturing.

The present model, however, is specific to the HfO_x ALD system, not representing a universally transferable predictor across different tools, facilities, or chemistries. Future improvements in transferability may be enabled by incorporating system-indicated descriptors that reflect chamber configuration and process environment, thereby supporting more robust digital twin frameworks for ALD process development.

Discussion

In summary, this study demonstrates that DNNs can effectively support ALD process optimization. By integrating WER as a metric for film density alongside T_{ox} and RI, we enabled a more comprehensive evaluation of thin-film characteristics. The DNN model achieved high predictive accuracy, exceeding 90% within ranges of ± 1.0 nm for T_{ox} , ± 0.04 for RI, and ± 0.9 Å/min for WER. Beyond quantitative predictive task, the generation of prediction maps provides a practical pipeline for incorporating ML into ALD workflows, laying the groundwork for prototype

Editorial Summary:

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Peer review information:

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